

Q1. Multiple-Choice Questions

Choose the most appropriate answer from the options provided.

(15 POINTS)

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
b	b	c	c	c	b	d	c	b	b	a	c	d	d	d

- Which cloud service model allows developers to build and deploy applications without worrying about underlying infrastructure?
 a) Infrastructure as a Service (IaaS) ☒ b) Platform as a Service (PaaS)
 c) Software as a Service (SaaS) d) Function as a Service (FaaS)
- Which of the following is a cloud deployment model that is specifically designed for a single organization and can be managed internally or externally?
 a) Public cloud ☒ b) Private cloud c) Hybrid cloud d) Community cloud
- Which cloud service model provides the highest level of abstraction and allows users to access applications over the internet without the need for installation or maintenance?
 a) Infrastructure as a Service (IaaS) b) Platform as a Service (PaaS)
☒ c) Software as a Service (SaaS) d) Function as a Service (FaaS)
- Which of the following is NOT a potential benefit of adopting cloud computing?
 a) Cost savings b) Improved scalability
☒ c) Reduced data security d) Increased flexibility
- Which of the following is a challenge in network virtualization?
 a) Resource overprovisioning.
 b) Limited scalability.
 c) Increased latency.
 d) Incompatible protocols.
- What is the primary benefit of using containers in a cloud environment?
 a) Improved physical hardware utilization
☒ b) Application isolation and portability across environments
 c) Faster physical server provisioning
 d) Enhanced network performance
- Which approach involves moving applications to the cloud without making significant changes to their architecture?
 a) Re-architecting b) Re-platforming
 c) Retain ☒ d) Lift-and-shift migration
- Which of the following technologies enables efficient network packet forwarding between virtual machines on the same physical server?
 a) VLAN b) Network Address Translation (NAT)
☒ c) Virtual switch d) Software-defined networking (SDN)

9. Which of the following is NOT a phase in the migration to the cloud?
- a) Assessment and planning
 - b) Data backup and recovery
 - c) Application migration
 - d) Testing and validation
10. Which of the following is a key characteristic of a baseline?
- a) It is a static configuration file used for system restoration
 - b) It represents a snapshot of normal system performance or behavior
 - c) It only applies to security configurations in a cloud environment
 - d) It eliminates the need for system monitoring tools
11. In which cloud service model is the customer responsible for managing the operating system and applications?
- a) IaaS
 - b) PaaS
 - c) SaaS
 - d) DBaaS
12. What is the purpose of a Virtual Private Cloud (VPC)?
- a) To enable serverless computing
 - b) To automate application deployment
 - c) To provide isolated cloud resources within a public cloud
 - d) To deploy virtual machines locally
13. The company has been using an antiquated billing software for more than 20 years longer than most of their staff has been employed. IT department has been asked to make a recommendation as to how to proceed with this phase of the migration. Given these conditions, what is the best approach to migrating the billing functions to the cloud?
- a) Retire
 - b) Rehost
 - c) Replatform
 - d) Replace
14. What is a key characteristic of software-defined networking (SDN)?
- a) Fixed network configurations
 - b) Hardware-based routing decisions
 - c) Limited scalability
 - d) Centralized control over network traffic
15. In a hybrid cloud setup, what is the cloud bursting strategy?
- a) Ensures consistent backup
 - b) Reduces latency in private networks
 - c) Simplifies disaster recovery
 - d) Offloading excess workloads to the public cloud

2. Write one of the unique differences for each of AWS, Azure and GCP in terms of cloud infrastructure? (3 POINTS)

3 AWS: has the most regions number of region, provide private networking, has VPC and it can span around ~~data centers~~ AZ on same region

Azure: has ~~the second~~ + 54 of regions, its virtual network called Vnet, it doesn't provide private network, Vnet can span around AZ on same region

GCP: has the least regions from them, it allow global accessing on its VPC, resources on GCP can communicate with each other whenever their region is

3. Compare and contrast public, private, and hybrid cloud infrastructures. What are the key differences? Discuss how organizations determine the most suitable cloud model based on their operational and security requirements. (4 POINTS)

~~public cloud: shared and flexible~~

4 public cloud: shared and flexible providing for services
* shared physical ^{server} ~~servers~~ ~~data centers~~ for multiple users or organization
* managed by the cloud service provider * less secure

private cloud: flexible providing for services

- * the company or organization own a physical ^{server} ~~data center~~
- * more secure
- * the organization or the CSP manage the server
- * more cost

hybrid cloud: ~~mixed but~~ mix their private and public cloud
flexible providing for services
allow company to achieve security and availability

Q2. Computational Questions (5 marks)

1. Adan is mapping out address pools for the subnets he plans to create in his Azure account. He has determined the network ID will use 192.169.100.0/24, and he needs 4 small subnets within this space. How many bits should Adan borrow from the host ID to create these subnets and what is the network ID of each subnet? (2 POINTS)

$$2^n \geq 4$$

$$n = 2$$

adan need to borrow two bits from host ID

Network ID = 192.169.100.0/26

① 192.169.100.64/26

② 192.169.100.128/26

2. Douglas is setting up VM instances on a Hyper-V server to host some basic web server. He needs to know how many vCPUs to allocate to each VM. The server's motherboard has four CPU sockets, each populated with eight-core processors, and hyperthreading is enabled. He's decided that the VMs' workloads can easily handle a 3:1 overcommitment ratio. He plans on creating 10 VMs at first and eventually increasing to 25 VMs as he fine-tunes the workloads and resource allocations. How many vCPUs should Douglas initially allocate to each VM? (3 POINTS)

~~# of threads = # of sockets * # of physical core * 2~~

3 sockets * 8 cores * 2 = 48 threads

3

8

2

~~# of threads = # of sockets * # of physical core * 2~~

Q3. Essay Questions (10 POINTS)

1. What is a hypervisor, and what are its types? (3 POINTS)

is a technology that we need to enable in order to create do virtual machines, it manage the process between the host and the guest machine

type 1: placed in the under layer & it provide virtualization when boot up

type 2: its an application that we run or download to create virtual machine

of threads = # of cpu socket * # of physical core * 2

$$= 4 * 8 * 2 = 64 \text{ threads}$$

3 : 1

59 : 64

number of vCPU = 192

should allocate 1 test it then add for each VM